RAB for NatureLM

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Abstract

Audio-language models (audio-LLMs) have become increasingly valuable for bioacoustic classification due to their ability to reason about audio and provide richer outputs than pure classification. However, converting audio into an LLM's token space leaves out discriminative information, and it remains unclear how to close the accuracy gap with traditional classifiers. We approach this by introducing retrieval-augmented generation (RAG) to NatureLM, an audio-LLM for bioacoustics, evaluating three retrieval strategies: BirdNET's native classifier output, BirdNET embeddings, and BEATs embeddings, using FAISS to extract relevant embeddings as additional textual context for the model to reason over. Across 770 test clips spanning 77 North American species, we find that RAG significantly improves classification accuracy over both NatureLM alone and standalone BirdNET classifier, and that retrieval source matters more than retrieval mechanism. BirdNET-based context consistently outperformes BEATs whether or not embedding or classification confidence was used. Additionally, we found that accuracy gains from additional context plateau beyond $k \approx 7$ candidate species, and that NatureLM was 5x more sensitive to textual information than auditory clues through modality ablations with random context and random audio. These findings suggests that the text-conditioning pathway, not just audio pathway, is a critical and underexplored topic for improving audio-LLM performance in bioacoustic classification.

1 Introduction

Automated bioacoustic classification has traditionally relied solely on audio classifiers, but the field has shifted towards large language model (LLM) based pipelines which can reason about audio and provide more information than pure classification. Although this change allows flexible responses, it often trades off with classification performance. The cost of converting bird audio to the token space of an LLM eliminates discriminative information that classifiers rely on. Thus, audio-LLM systems typically underperform their traditional classifier counterparts in accuracy. This disparity is especially prevalent for rare or acoustically similar species with few training examples.

One way to overcome this loss of information is to give the LLM additional context at inference time rather than relying on the audio encoder. This new system, called RAG (retrieval-augmented generation) [Lewis et al., 2020], has been effective at giving LLM outputs external knowledge for

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text-based responses; however, it has only recently been introduced in audio-conditioned bioacoustic classification. Related studies have explored how free-text metadata can improve classification accuracy through informative context [Robinson et al., 2025], which enhances retrieval pipelines [Maiti, 2025], but none have directly explored how retrieval-augmented context affects an audio-LLM's classification accuracy or which retrieval strategy works best based on model sensitivity to textual information.

In this work, we study how to design RAG for audio-LLMs, using NatureLM as our baseline and North American bird vocalizations as our data. We build a RAG pipeline that retrieves candidate species using BirdNET embeddings, classifiers, or general-purpose BEATs embeddings, which we use as context for the NatureLM prompt. Using 770 test clips across 77 species, we evaluate which retrieval strategy improves classification accuracy the most, how much context (k) is optimal before accuracy diminishes, and how much the model depends on the retrieved context versus the audio itself, using controlled tests that scramble context and audio.

Our results show that retrieval-augmented context significantly improves NatureLM classification accuracy over both a standalone NatureLM model and a BirdNET classifier. We find that a fine-tuned acoustic classifier (BirdNET) gives substantially better retrieval context than a general-purpose audio encoder (BEATs), and that accuracy gains are less meaningful beyond k $\approx$ 7 candidate species. Our experiments further show that NatureLM is more sensitive to the validity of retrieved textual context rather than audio input, which highlights the value of text-conditioned pathways, which are often unexplored yet critical for improving audio-LLM bioacoustic classification.

Our contributions are:

1. We introduce and evaluate a RAG framework for audio-conditioned bioacoustic classification, the first (to our knowledge) to provide quantitative, statistically validated comparisons across various retrieval strategies.

2. We identify BirdNET-based context as the most effective retrieval strategy, and show that it is statistically significant and beats general-purpose audio embeddings.

3. We quantify the context-size tradeoff, showing that classification accuracy gives non-significant returns beyond $k \approx 7$.

4. Through audio/context ablations, we find NatureLM relies roughly 5x more heavily on quality of retrieved textual context than audio information.

2 Related Works

The Earth Species Project introduced NatureLM [Robinson et al., 2025], a bioacoustic foundation model that converts BEATs audio embeddings into Q-Former tokens that can be digested by an LLM's LoRA-adapted weights; this model is then evaluated using zero-shot classification and detection benchmarks. Their evaluation format gives the model a candidate species list as part of the prompt (like a multiple-choice question) with an answer key. This is a good benchmarking method, but they don't reflect real-world deployment, where the correct answer isn't known in advance. Additionally, they also underscore that free-text notes in the prompt improve accuracy by 2.2 points, showing that textual context plays a role in improving accuracy. We build on this experiment by using NatureLM's architecture but generalize their candidate-list setup using a dynamic retrieval mechanism. While their study uses Levenshtein distance postprocessing to narrow results, we focus on the text-conditioned pathway before the LLM's decision-making, which their paper doesn't explore.

RAB proposes a retrieval-augmented pipeline for bioacoustics [Maiti, 2025]: audio embeddings are indexed into a library, new queries retrieve nearest neighbors and metadata, and this evidence informs a generative part. However, their experiment is not fully developed or evaluated for generative purposes, and they have no quantitative results. We implement and evaluate RAG context directly within an audio-LLM classification pipeline, providing (to our knowledge) the first quantitative, statistically validated comparison across multiple retrieval strategies for this setting.

Teramota and Kojima created a multimodal bird song RAG system with a slightly different task and similar architecture [Teramoto and Kojima, 2025]. Their product requires a user query with a combination of Wikidata, which allows the system to retrieve a matching species and generate a response. There are multiple embedding types: graph relations between species, names, and audio.

Their RAG differs from our system because their retrieval returns a single species entity, while the generation is an LLM turning that response into a description. The LLM makes no classification because the vector search already grabs the correct species. Furthermore, their classification layer is done during the extraction of the embeddings rather than with the LLM. Through experimentation, they find multimodal combinations improve model classification performance when compared to audio embeddings alone. However, too many modalities (all three in their case) actually underperform the best two-modality combination, which emphasizes the value in a balance between noise and meaningful context. Our architecture builds on this idea but works differently, finding the right amount of retrieval-augmented context for NatureLM to avoid plateauing the results or hurting the model accuracy.

Additionally, SpeechT-RAG addresses the challenge of applying LLM-based classification to audio-derived tasks, where converting acoustic signals into text for a text-only LLM discards information the model would need to accurately classify [Zhang et al., 2025]. Rather than relying on audio-conditioned generation, they propose retrieving relevant textual examples at inference time to give the LLM additional context. This experiment is similar to ours because it shows that text-only LLM performance is insufficient for audio-derived tasks and therefore incorporates RAG to provide relevant examples during inference rather than relying solely on fine-tuning. Zhang's study highlights how generic retrieval can even degrade performance, which is why RAG strategies for acoustic signals need to be specifically designed for the task. Zhang only analyzes text to output an answer, but we maintain audio-conditioned generation with context from text-based RAG. Their design works well when the acoustic patterns are well understood and can be captured by a predefined feature set like binary depression detection. However, NatureLM, which we use, does not require setting acoustic statistics in advance. With bioacoustics specifically, and especially with rare species, having a predefined set is extremely difficult to achieve, which means that their prototype isn't dynamic enough for the given task at hand. Therefore, while we take some elements of RAG for this experiment, we use NatureLM's architecture, which still maintains audio-conditioned generation.

Regarding our embedding strategy, we used BirdNET, which is a deep convolutional model with 27 million parameters. Kahl et al. trained this model to classify 984 North American and European bird species from short audio clips, using mel-spectrogram inputs and domain-specific data augmentation [Kahl et al., 2021]. It was trained on recordings from Xeno-canto and the Macaulay Library and evaluated across single-species clips. Their experiment on clean focal recordings achieved a mAP of 0.791 but had a substantial drop on real-world noisy soundscape data F0.5: 0.414, which shows a negative correlation between noise and accuracy. Additionally, Kahl et al. mention that rarity of species alone isn't the hardest to classify but rather species that closely mimic other species' calls. We use BirdNET directly as a retrieval component in two of our RAG strategies through its native softmax classification output for candidate ranking and its embedding space for FAISS nearest-neighbor retrieval. Instead of using BirdNET as a standalone classifier, we study it as a context-providing retriever feeding into a separate generative model; through this, we find that BirdNET-derived context outperforms general-purpose audio encoders, which aligns with Kahl et al.'s notion that domain-specific model design outperforms generic architectures.

Large pre-trained models store substantial factual knowledge in parameters, but this knowledge is difficult to update after training. Lewis et al. discuss how models with a differentiable access mechanism can overcome this issue through reality-augmented generation, which uses a combination of a parametric pre-trained model with a non-parametric retrieval mechanism [Lewis et al., 2020]. They found RAG outperformed simple parametric models on knowledge-intensive tasks such as open-domain QA benchmarks. RAG also produced more specific and less repetitive text than a parametric-only model. We use Lewis et al.'s architecture of RAG alongside NatureLM, which acts as the parametric component that generates the prediction. Our experiment lies in the non-parametric component, figuring out the best retrieval strategy, context, and how much RAG matters using similar architectures to this study.

Additionally, Li et al. notice a big cost associated with training large-scale models, so they propose BLIP-2, an effective pre-training strategy that connects frozen image encoders to frozen pretrained LLMs without having to fine-tune either end-to-end [Li et al., 2023]. This problem solves a modality mismatch, as an image encoder produces a large number of dense visual embeddings but LLMs expect a much smaller number of inputs. Q-Former bridges these two worlds together through a small set of learnable query vectors that attend over the encoder's output to figure out the most useful features to feed into the LLM. This architecture is important because it resembles NatureLM's internal architecture, using Q-Former to convert audio into an LLM representation. However, we provide an additional layer of RAG on the text-conditioning side, which enhances the overall pipeline.

Regarding LLM's, fine-tuning billions of parameters has become extremely costly and slow. For example, GPT-3 has over 175B parameters [Hu et al., 2021]. Hu et al. discuss Low-Rank Adaptation (LoRA) which freezes the pre-trained model weights and injects trainable rank decomposition matrices inside layers in the main model to fine-tune a smaller subsection of the model. Through such a design, they assert that LoRA can reduce the number of trainable parameters by 10,000 times and GPU memory requirements by 3 times. Our NatureLM model uses LoRA as part of the training process by freezing all of LlaMA's original weights to create NatureLM's base architecture. This is important to fine-tune the model for interpreting audio tokens without modifying LLaMA's original capabilities while also avoiding high computational cost.

Similarity searches with embeddings are especially important where systems have to deal with complex data like audio, but they require a lot of computation. Johnson et al. propose a k-selection for nearest neighbors, Facebook AI Similarity Search, that is 8.5x faster than a prior GPU state of the art [Johnson et al., 2017]. This model is built for GPUs, which excel at parallel execution and capitalize on their memory hierarchy for more efficiency. Additionally, FAISS uses approximate search instead of always doing exact search, such as product quantization. This model was used in our RAG experimentation for deriving embeddings to be used as context for the model.

Bird classification involves multiple model types and architectures that deal with large data processing and many parameters. Perch is another large-scale bird classification model which is similar to BirdNET but trained with a focus on higher quality feature embeddings for few-shot transfer learning [Ghani et al., 2023]. Their findings showed that Perch consistently outperformed embeddings from general-purpose audio models trained on non-bioacoustic data. Additionally, Perch and BirdNET achieve similar classification performance but Perch was more consistent across wider range downstream transfer tasks. Both Perch and BirdNET were trained diretly on datasets that overlap with NatureLM, the model we use. However, neither Perch or BirdNET incorporate retrieval-augmented context or LLM-based generation, which is the gap we address by using these classifiers' outputs as retrieval signals for audio LLMs.

Audio-MAE (Masked Autoencoders that Listen) is a self-supervised audio representation learning method which adapts masked autoencoding to audio spectrograms [Huang et al., 2022]. Audio-MAE involves masking a large portion of the spectrogram patch, encoding the unmasked patches, and training a decoder to reconstruct missing portions from this partial context. The encoder learns general-purpose audio representations without requiring labeled data. BirdNET and Perch are both trained directly for labeled classification tasks, while AudioMAE is domain-agnositc, which is trained on general AudioSet data. Ghani et al. [Ghani et al., 2023] use this finding to compare AudioMAE's embeddings against BirdNET and Perch for bioacoustic transfer learning and find that AudioMAE underperforms both bird-specific architectures, which highlights the value of encoders being fine-tuned for specific domains. This aligns with our experiment because we compare BirdNET's RAG capabilities to BEATs, which is a general-purpose audio model, and find that domain-specific context outperforms BEATs.

CLAP (Contrastive Language-Audio Pretraining) models learn a joint embedding space combining audio and natural language through an audio and text encoder that pulls matching audio-text pairs closer in embedding space and pushes mismatches away [Elizalde et al., 2022]. This allows for an informative general-purpose audio-text representation which supports zero-shot audio classification without needing task-specific fine-tuning. While CLAP isn't designed for species-level classification because it is trained on general-purpose audio-text pairs, its retrieval-style between audio and text conceptually relates to our BirdNET and BEATs embedding space strategy. However, in our experiment, our retrieved context is not compared to the query in a shared audio-text embedding space because retrieval candidates become textual context for a separate generative model to condition on, rather than a similarity score for direct classification.

McNemar's test is a statistical test for paired data, used to determine whether two related classifiers or conditions differ significantly in their error rates on the same set of examples [McNemar, 1947]. It evaluates the cases in which the two conditions disagree, which makes it well-suited to compare paired predictions on an identical test set. We use this in our experiment to compare retrieval strategies evaluated on the same 770 clips.

Bootstrap is a resampling method for estimating the sampling distribution of a statistic by repeatedly resampling the observed data with replacement [Efron, 1979]. We use this approach to compute 95% confidence intervals for accuracy across all reported conditions, resampling our 770 test clips 10,000 times with replacement.

Liu et al. study how well large language models use information across long input contexts, testing on tasks like multi-document question answering, where the correct answer is hidden among other distractor documents in different positions [Liu et al., 2023]. They find that model performance is highest when relevant information sits at the beginning or end of context, and gets worse when it sits in the middle, which produces a U-shaped accuracy curve. More importantly, they find that performance can degrade as the number of documents in context grows, even if the correct information is present. This aligns with our k-sweep findings, where accuracy gains beyond 7 species start to produce less meaningful results because it adds noise to the reasoning LLM. While Liu et al. study general-purpose text-only LLMs and document position, our results suggest a related but distinct phenomenon in the audio-LLM setting: NatureLM does not merely fail to benefit from additional context past a certain point, but is measurably harmed by the presence of low-quality or irrelevant candidates, regardless of position.

Large Language models are especially good at natural language processing tasks. Yet, these performances are dependent on the quality of input context for solving a task. Shi et al. investigate the distractibility of LLMs in Grade-School Math with Irrelevant Context [Shi et al., 2023]. They find that a single piece of irrelevant information can significantly degrade model performance; furthermore, explicitly instructing the model to ignore irrelevant information can minimize the reduction in accuracy. This mirrors the findings we have through our random-context ablation: confusing textual context for the model significantly decreased accuracy regardless of prompting methods. This reinforces the idea that RAG pipelines must prioritize retrieval precision over recall, because context significantly affects LLM accuracy. While Shi et al. study distraction from single inserted irrelevant sentences, we extend this concern to a multimodal retrieval-based setting.

Self-supervised learning has been used in AI to create its own models in the audio domain; however, audio contains a lot of low-level detail that doesn't benefit understanding, which leads to reconstruction loss. Chen et al. propose BEATs, which instead trains the model to predict discrete labels after feeding the audio through a separate acoustic tokenizer [Chen et al., 2022]. The tokenizer and the model are jointly refined in training through iteration as well. NatureLM uses BEATs in its architecture because it is designed to produce more semantically meaningful embeddings, which is more useful as an input that the Q-Former compresses for the LLM to reason over.

3 Methodoloy

3.1 System Overview

The current system for bioacoustic classification is NatureLM, an audio-language model built on LLaMA 3.1 8B Instruct as its underlying LLM. Figure 1 shows the overall pipeline, including our retrieval-augmented generation (RAG) layer.

Our base system is NatureLM, an audio-language model built on Llama 3.1 8B Instruct, which takes raw audio and outputs a predicted species name [Robinson et al., 2025]. Input audio is processed by a BEATs encoder [Chen et al., 2022], which produces a sequence of dense embeddings that capture acoustic features in the recording. These embeddings however are not understood by the LLM; so, a Q-Former [Li et al., 2023] module compresses the embeddings into a fixed number of learned tokens that are in the same representation space as LlaMA's text tokens. Additionally, LlaMA is not natively trained to interpret audio-shaped tokens, so NatureLM incorporates LoRA adapters [Hu et al., 2021] which teach the model to associate audio tokens with species identity and behavior rather than noise.

NatureLM [Robinson et al., 2025] uses a tunable α parameter ($\alpha \in [0, 1]$ that trades off classification accuracy for conversational skills: $\alpha = 1$ favors zero-shot classification accuracy with minimal conversational skills. $\alpha = 0$,on the other hand, favors conversational flexibility at the cost of classification performance. All experiments use $\alpha = 0.5$ as a fixed operating point.

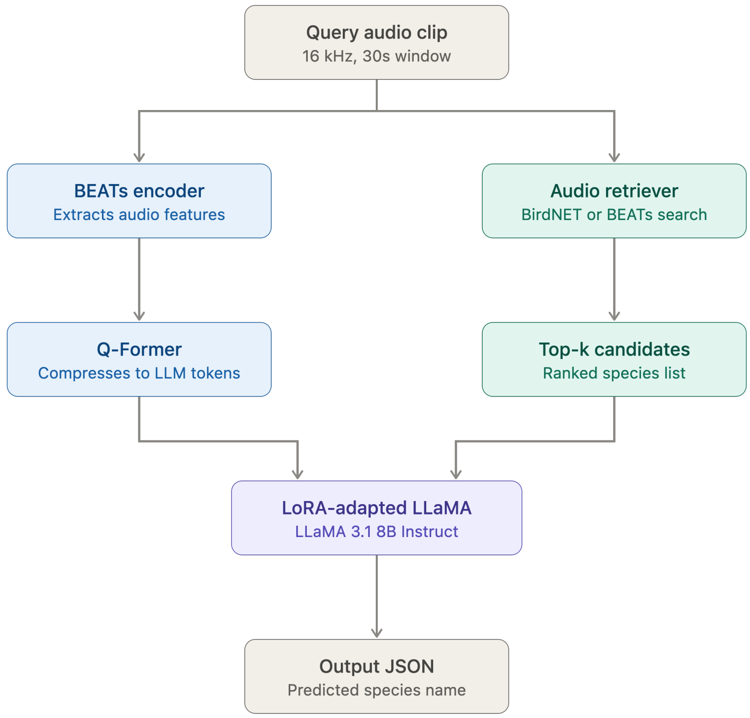

Figure 1: NatureLM RAG architecture. Query audio is processed by both the existing audio pathway (BEATs encoder, Q-Former) and the retrieval pathway introduced in this work (audio retriever, top-k candidate list), which are merged inside the LoRA-adapted LLaMA backbone to produce a structured output.

3.2 Retrieval-Augmented Generation Layer

In this experiment, we introduce a retrieval-augmented generation (RAG) layer that gives additional text content to NatureLM before generation. RAG doesn't alter the audio pathway; it takes the audio clip, independently processes the information, and returns a ranked list of the top-k candidate species [Lewis et al., 2020]. We evaluate three retrieval strategies:

- **BirdNET classifier RAG**: retrieves the top-k species using BirdNET's native softmax classification output, ranked by confidence.
- **BirdNET embedding RAG**: retrieves the top-k nearest neighbors in BirdNET's embedding space using FAISS, based on similarity to the query clip's embedding.
- **BEATs embedding RAG**: retrieves the top- k nearest neighbors using FAISS over embeddings from BEATs, a general-purpose audio encoder.

The first strategy uses a domain-specific classifier's confidence ranking, while the other two use FAISS retrieval over a pre-indexed embedding space. BirdNET's embedding space is fine-tuned for bird acoustics, while BEATs is general-purpose. This design allows us to test the effect of *retrieval mechanism* (classification vs. embedding similarity) and *representation source* (domain-specific vs. general-purpose).

3.3 Prompting

Retrieved candidates are formatted as a ranked list and added to the instruction prompt, which is given to NatureLM. For example:

```
<retrieved_context>
BirdNET classification predictions (by confidence):
  1. American Redstart (confidence=0.842)
  2. Canada Warbler (confidence=0.341)
  3. Black-and-white Warbler (confidence=0.187)

What species is most likely vocalizing in this recording?
Respond ONLY with valid JSON and nothing else: {"common_name": "...", "scientific_name": "..."}
```

3.4 Data

We construct the evaluation set from Xeno-Canto recordings using the "no quality" filter, restricted to recordings uploaded on or after January 1, 2024, in order to reduce the likelihood that clips were seen during training by NatureLM or BirdNET [xen, 2024]. We manually curated a candidate list of 100 North American bird species that optimized acoustic difficulty (rare species or similar to other species in the set) and common availability. After shuffling clips with a fixed random seed and filtering species with at least 20 available recordings, our final dataset consists of 77 species with 20 clips each, split evenly into 10 reference clips (used to build the retrieval index) and 10 test clips per species (770 test clips total).

3.5 Audio Preprocessing

Audio clips were downloaded from Xeno-Canto [xen, 2024] with a maximum duration of 60 seconds (mean: 24.1s across the final test set). NatureLM's [Robinson et al., 2025] pipeline defaults to a 10-second processing window, which would discard most of each recording given our clips' mean duration of 24.4s. We instead configure the pipeline to use a 30-second window, allowing the majority of clips to be processed in a single forward pass rather than being truncated or split. Clips shorter than 30 seconds are zero-padded internally by NatureLM; clips exceeding 30 seconds (up to our 60-second maximum) are split into two non-overlapping windows, of which only the first is used for prediction. In contrast, our BirdNET [Kahl et al., 2021] and BEATs [Chen et al., 2022] retrieval components process representations over the full clip regardless of length, since both support variable-length inputs natively. Thus, for the majority of clips (65%, $\leq$30s), NatureLM and the retrievers operate on identical audio content; for the remaining 34% (30–60s), retrieved context may reflect audio beyond what NatureLM directly evaluates. We assess the impact of this discrepancy in Results (Section 4.4) and discuss it further in Limitations.

3.6 Statistical Analysis

We report raw classification accuracy as the proportion of exact species-name matches across 770 test clips. To understand if differences between conditions are statistically meaningful, we use McNemar's test [McNemar, 1947] on paired disagreements between conditions, and report 95% confidence intervals via bootstrap resampling [Efron, 1979] (10,000 samples with replacement, 2.5th/97.5th percentile bounds).

To understand the role of context size, we repeat the RAG experiment across $k \in 1, 2, 3, 4, 5, 7, 10, 12, 15, 20$ retrieved candidates. To isolate the contribution between audio and test, we run two ablations: (1) random context, where the correct audio clip is paired with a randomly sampled candidate list, and (2) random audio, in which the correct candidate list is paired with a randomly sampled audio clip. Both ablations are repeated with the same range of k values as the main experiment. [1]

Finally, to assess if the window-length mismatch between clips introduces systematic bias, we compare accuracy between clips $<=$ 30s and clips 30-60s using a χ^2 test of independence for each condition [Pearson, 1900].

[1] Random substitution for the ablation conditions excludes only the exact source clip, not other clips of the same species. Given that 769 clips remain, with 9 belonging to the same species, yields an ≈1.2% chance per trial that the randomized context or audio clip accidentally corresponds to the correct species. Across all 770 evaluations, this implies an expected upward bias of roughly 9 clips or ≈1.2% in reported ablation accuracy, which is small relative to the 42-54% and 6.5-11.6% change effects observed, and doesn't change our conclusion.

4 Results

We report raw accuracy based on an exact-match agreement between the model's predicted species name (JSON format) and ground truth, across 770 test clips spanning 77 species. Stastical significance between findings is asssessed through McNemar's test [McNemar, 1947] on paired disagreements, and 95% confidence intervals are computed using bootstrap resampling [Efron, 1979] (10,000 resamples with replacement)

4.1 Retrieval strategy comparison

Table 1 summarizes raw accuracy across baseline and three retrieval strategies

Condition	Accuracy
NatureLM alone (α=0.5)	58.7%
BirdNET classifier	62.5%
RAG (BirdNET embedding)	68.2%
RAG (BirdNET classifier)	66.8%
RAG (BEATs embedding)	59.1%

Table 1: Retrieval strategy comparison

All RAG variants were compared against the NatureLM baseline and against each other using McNemar's test (Table 2).

Comparison	Δ Accuracy	p-value
BirdNET cls. vs. NatureLM alone	+3.8%	0.0385*
RAG (BirdNET cls.) vs. BirdNET cls.	+4.3%	0.0000***
RAG (BirdNET cls.) vs. NatureLM alone	+8.1%	0.0000***
RAG (BirdNET emb.) vs. NatureLM alone	+9.5%	0.0000***
RAG (BirdNET emb.) vs. RAG (BirdNET cls.)	-1.4%	0.4277 n/s

Table 2: McNemar's test results across retrieval conditions

Both BirdNET retrieval strategies (classifier and embedding) significantly outperform NatureLM alone and the standalone BirdNET classifier. Additionally, the difference between the two BirdNET-based retrieval strategies (classifier vs embedding) is *not statistically* significant, yet both outperform BEATs-based retrieval. This shows that the deciding factor may not be from the retrieval mechanism (classifier score vs nearest neighbor embedding) but the retrieval source. BirdNET is fine-tuned specifically for bird acoustics, so it produces more useful context than BEATs, a general-purpose audio encoder, regardless of how that context is retrieved. BEATs-based RAG is statistically not significant when compared to the no-RAG baseline, implying that retrieval only helps when the information used in the context is task-relevant.

4.2 Context size

After understanding the retrieval strategies, we run the same experiment on the BirdNET classifier but vary the number of retrieved candidate species (k) provided in the prompt.

With the BirdNET classifier, we see that accuracy improves significantly from $k = 1$ to $k = 4$ (+2.2%, p = 0.0171) and continues to improve slowly through $k = 20$, which is a 3.4% increase relative to $k = 1$ (p = 0.00007). However, the pairwise comparisons among k values above $k = 7$ show no significant difference. This indicates that the benefit of additional context to provide the right species diminishes around $k = 7$, where future gains are not distinguishable from noise. This means that NatureLM does not require a large candidate list to benefit from retrieval; smaller, well-chosen candidates (4-7) provide most of the achievable improvement, wheras $k = 1$ and $k = 20$ have only 3.7% improvement.

k	Accuracy	95% CI
1	63.9%	[60.5%, 67.3%]
2	65.5%	[62.1%, 68.8%]
4	66.1%	[62.9%, 69.5%]
5	65.8%	[62.5%, 69.2%]
7	65.8%	[62.5%, 69.2%]
10	66.5%	[63.2%, 69.7%]
12	66.8%	[63.4%, 70.0%]
15	66.9%	[63.5%, 70.3%]
20	67.3%	[63.9%, 70.5%]

Table 3: Accuracy of BirdNET classifier RAG across context sizes ($n = 770$ per condition).

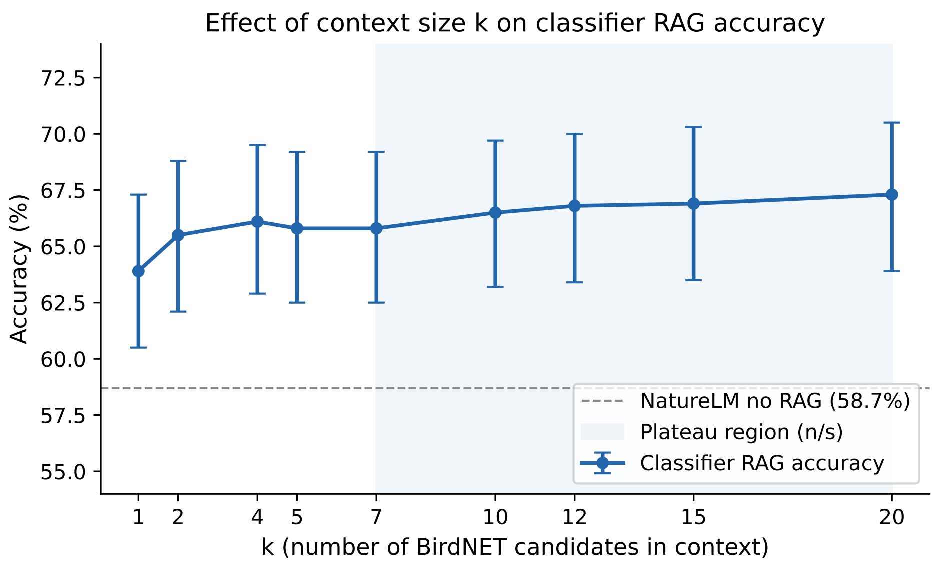

Figure 2: Effect of context size k on BirdNET classifier RAG accuracy. Accuracy gains plateau beyond $k \approx 7$ candidates (shaded region), with no statistically significant difference between k=7 and k=20.

4.3 Ablations: context vs. audio sensitivity

As a reference point, we also compute a theoretical chance-level floor of 1.3%, which is calculated by randomly getting 1/77 species right. The empirical null (8.6%) seen in the image below reflects the expected accuracy if NatureLM relies on copies from the provided context, which is chosen at random. Since random_ctx draws 7 candidate species at random from the pool, there's a 7/77 $\approx$ 9.1% chance the correct species is included, which closely matches the observed 8.6%. This floor is calculated from random species and random confidence scores as context.

To figure out how much NatureLM relies on this retrieved context vs. the audio signal, we ran two ablations. Random context (correct audio paired with a randomly selected but incorrect candidate list) and random audio (correct context paired with a randomly selected but incorrect audio clip) were repeated across k values.

Compared to the regular-context condition for each k, both ablations perform significantly worse (all $p < 0.0001$).

Additionally, the magnitude the of drop differs sharply between the two ablations: correct audio with wrong context drops accuracy by 42-54 percentage points across k values, while corrupting the audio with correct context only drops accuracy by 6.5-11.6 percentage points. This difference in percentage changes (about 5 times more for random context) indicates that NatureLM's classification decisions are substantially more sensitive to the quality of retrieved context rather than the audio signal.

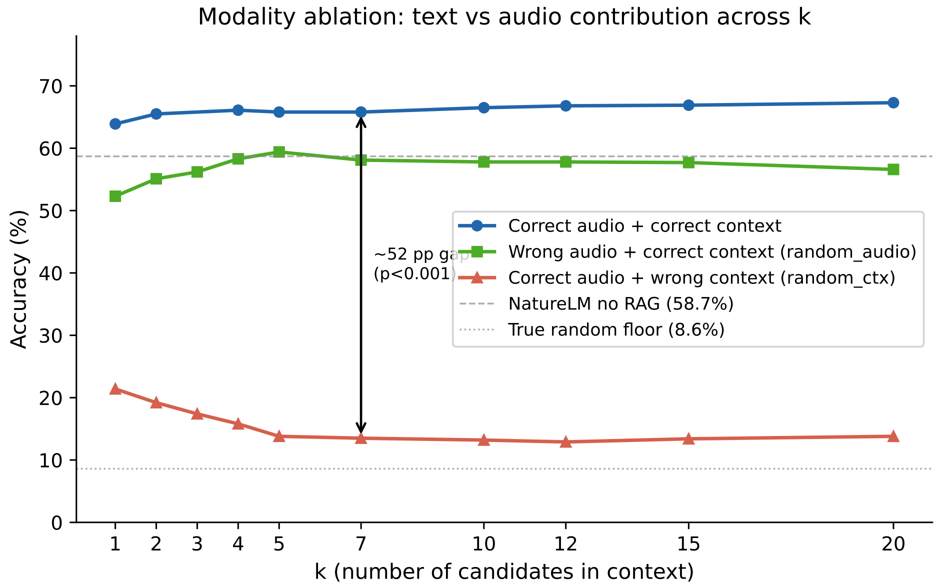

Figure 3: Modality ablation across context sizes k. Corrupting the retrieved context (red) causes a substantially larger accuracy drop than corrupting the audio input (green), relative to the regular condition (blue), indicating NatureLM's classification decisions are more sensitive to context quality than to audio.

k	Accuracy	95% CI
1	21.4%	[18.4%, 24.4%]
2	19.2%	[16.5%, 22.1%]
3	17.4%	[14.8%, 20.1%]
4	15.8%	[13.4%, 18.4%]
5	13.8%	[11.3%, 16.2%]
7	13.5%	[11.2%, 16.0%]
10	13.2%	[10.9%, 15.7%]
12	12.9%	[10.5%, 15.3%]
15	13.4%	[11.0%, 15.8%]
20	13.8%	[11.3%, 16.2%]

Table 4: Accuracy under randomized (incorrect) context, correct audio retained.

k	Accuracy	95% CI
1	52.3%	[48.8%, 55.8%]
2	55.1%	[51.7%, 58.6%]
3	56.2%	[52.7%, 59.7%]
4	58.3%	[54.8%, 61.8%]
5	59.4%	[55.8%, 62.9%]
7	58.1%	[54.5%, 61.6%]
10	57.8%	[54.4%, 61.3%]
12	57.8%	[54.3%, 61.3%]
15	57.7%	[54.2%, 61.2%]
20	56.6%	[53.1%, 60.1%]

Table 5: Accuracy under randomized (incorrect) audio, correct context retained.

k	Regular → Random Context	Regular → Random Audio
1	−42.5%***	−11.6%***
2	−46.2%***	−10.4%***
4	−50.3%***	−7.8%***
5	−52.1%***	−6.5%***
7	−52.3%***	−7.8%***
10	−53.2%***	−8.7%***
12	−53.9%***	−9.0%***
15	−53.5%***	−9.2%***
20	−53.5%***	−10.6%***

Table 6: Accuracy change relative to the matched regular condition. All comparisons $p < 0.0001$, $n = 770$ clips.

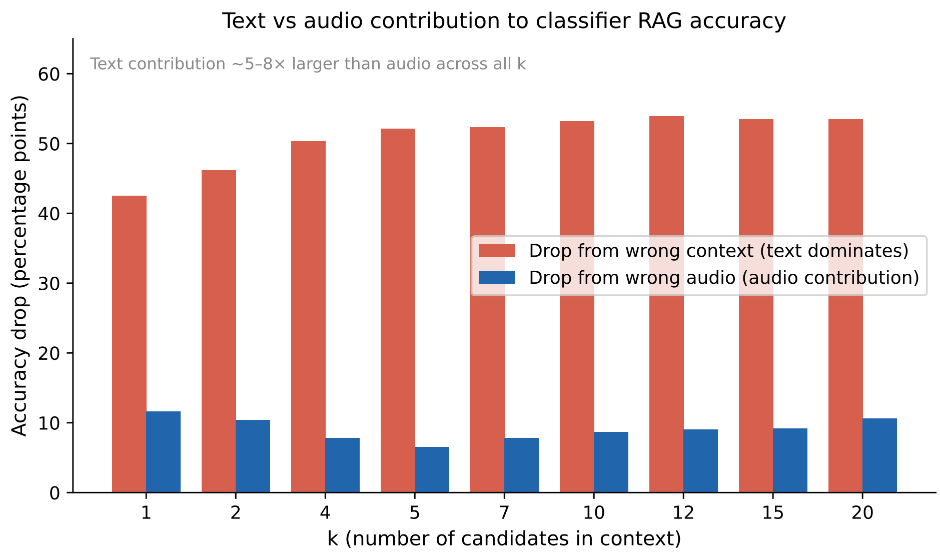

Figure 4: Accuracy drop from corrupting context vs. corrupting audio, shown as percentage points relative to the regular condition, across context sizes k. The drop from wrong context (red) is consistently 5–8× larger than the drop from wrong audio (blue), indicating text contribution dominates NatureLM's classification decisions.

We observe that random-context accuracy decreases as k increases (21.4% at k=1 to 13% after k=7), which is the opposite of the trend we saw with correct context. This aligns with our interpretations because additional incorrect candidates would add confusing noise rather than useful signals, which reinforces the idea that context quality rather than quantity drives performance.

4.4 Robustness to Window-Length Mismatch

Because we configured NatureLM to process only the first 30-second window of each clip while BirdNET and BEATs process over the full clip (Section 3.5), we determine whether this discrepancy introduces systematic bias by comparing accuracy between clips ≤30s (65% of the test set) and clips 30–60s (34%) using a χ^2 test of independence.

While there is a concern that window mismatch would harm performance, clips exceeding NatureLM's 30-second window show significantly *higher* accuracy under both BirdNET-based RAG conditions, while the NatureLM-alone baseline and BEATs RAG show a similar but non-significant trend. This pattern shows the window mismatch is not detrimental to retrieval quality. One plausible explanation is that BirdNET's aggregation over the full clip yields more reliable candidate rankings when more acoustic evidence is available for the model to make decisions; however, we cannot rule out that longer recordings are systematically easier to classify for other reasons (there could be clearer or

Condition	≤30s	30–60s	p (χ^2)
NatureLM alone	56.4%	63.1%	0.0727 n/s
RAG (BirdNET classifier)	62.4%	72.4%	0.0051**
RAG (BirdNET embedding)	64.9%	74.3%	0.0082**
RAG (BEATs embedding)	56.6%	63.8%	0.0519 n/s

Table 7: Accuracy by clip duration subset. All conditions $n = 770$ total (502 ≤30s, 268 30–60s).

more repeated vocalizations) independent of RAG, given the similar non-significant trend observed in the no-RAG baseline. We discuss this further in Limitations.

5 Discussion

Through multiple ablations and conditions for RAG, we find that retrieval source matters more than retrieval mechanism. There isn't a significant difference between BirdNET classifier RAG and BirdNET embedding RAG, but both outperform BEATs-based RAG. This suggests that how candidates are retrieved doesn't matter; rather, which representation they're retrieved from does. Zhang et al [Zhang et al., 2025] furthers this idea by noting that generic retrieval degraded performance, where domain-specific encoding is more important. We see that this same pattern for bird acoustics gives far better context than a general-purpose audio encoder, regardless of the retrieval mechanism. This idea is further supported by Kahl et al. [Kahl et al., 2021] as their BirdNET-specific CNN classifier outperformed general-purpose strategies. Additionally, they showed that the most difficult species to classify are the ones that mimic others, which is why it's important to have correct embeddings that point to the right candidate species; this is why text representations need to have a strong foundation for retrieval, and that comes with bringing domain-tuned systems.

More context is not always better context. Our k-sweep shows that accuracy gains plateau by $k \approx 7$ with no difference between $k = 7$ and $k = 20$. This parallels Teramoto et al.'s finding [Teramoto and Kojima, 2025] that combining all three of their modalities underperformed the best two-modality combination. In both cases, extra context started to harm the model's ability to classify because it provided confusing noise. Liu et al. [Liu et al., 2023] similarly find that LLM performance can degrade as the number of documents in context grows, even when the correct information remains present among them, suggesting that our k-plateau reflects a broader tendency of LLMs to be overwhelmed by excess context rather than a phenomenon unique to bioacoustic retrieval. Our random-context ablation reinforces this idea because accuracy under corrupted context actually decreases as k increases (21.4% at $k = 1$ and down to 13% by $k = 7$), which is consistent with more wrong candidates adding more confusion to the model. This notion suggests RAG for bioacoustics should favor a small set of high-quality candidates over an extensive list. Maiti [Maiti, 2025] raises a related design consideration in the theoretical RAB framework, proposing that generated outputs should explicitly cite the retrieved evidence behind them (e.g., timestamps, spectrogram snippets) to allow domain experts to verify predictions, and that retrieval depth could adaptively scale with input signal quality, though this adaptive-k idea is presented as a practical engineering choice rather than an evaluated methodological contribution.

NatureLM is far more sensitive to context than audio. Our results show that there is roughly a 5:1 asymmetry between corrupting context vs. corrupting audio. This means that corrupting context heavily decreases accuracy for NatureLM, which echoes a broader concern that generic or poorly designed retrieval heavily hurts performance in acoustic tasks [Zhang et al., 2025]. Shi et al. [Shi et al., 2023] finds this phenomenon common with LLMs, noting that a single piece of incorrect information degrades reasoning performance in Codex and GPT-3.5. NatureLM leans so heavily on retrieved context that low-quality retrieval could dominate the output even if the audio is informative. This means that text-conditioned pathways, not just audio pathways, are an underexplored level for improving audio-LLM classification.

Window-length mismatch does not appear to hurt results. As our NatureLM configuration only sees a 30-second window while retrievers aggregate over the full clip, we found that BirdNET-based RAG performs better on clips over 30 seconds. A plausible reason for this is that there is more acoustic evidence available to the retriever, but it is unknown whether the informative features came in the beginning or latter half of the clip. Our experiment confirmed that there was no systematic bias that brought RAG downward because it saw only 30-second clips, which held our main claims regardless.

6 Limitations

Single base model. All of our experiments use NatureLM, so findings about retrieval strategy, context size, and context sensitivity may not generalize to other audio-LLM architectures with different audio-conditioning mechanisms.

Our evaluation is restricted to 77 North American bird species chosen for common availability and acoustic difficulty. These results may not transfer to other bioacoustic domains or species with different call structures.

Scoring predictions by exact string match penalizes semantically correct but differently formatted predictions (other common names or historical names). This can add some noise to the raw accuracy numbers, though trends across conditions are unaffected since scoring is applied uniformly.

We configure NatureLM to use a 30-second processing window, an increase from the pipeline's 10-second default, chosen specifically to allow the majority of our clips (mean duration 24.4s) to be processed in a single forward pass rather than being aggressively truncated. However, for the 34% of clips exceeding 30 seconds (up to our 60-second maximum), only the first window is used for prediction, while our retrieval components (BirdNET, BEATs) process the full clip regardless of length. We directly tested whether this discrepancy disadvantages longer clips (Section 4.4) and found no evidence that it does; if anything, RAG performance is significantly higher on clips exceeding 30 seconds. Future work could extend NatureLM's window further, or aggregate predictions across multiple windows, to fully eliminate this discrepancy.

The random context and random audio ablations exclude only the exact source clip when sampling a replacement. However, other clips of the same species could be tested. With 769 remaining candidates per draw (9 of the same species remain), each draw has a 1.2% chance of accidentally giving the correct species for context, so this could lead to a 1.2% bias in percentage points. Yet, this is small relative to the gap between random context and random audio that was observed, so it doesn't change our conclusions.

All experiments use NatureLM's alpha parameter fixed at 0.5. Since alpha trades off classification accuracy against conversationality, RAG's relative benefit may differ at other settings and should be studied as future work.

7 Conclusion

This paper set out to build a RAG framework on top of NatureLM to understand how retrieval-augmented context affects bioacoustic classification of North American birds. We had four major findings. (1) RAG genuinely improves accuracy over both NatureLM alone and a standalone BirdNET classifier. (2) The source of retrieval matters more than the mechanism itself. We found that BirdNET beats BEATs regardless of whether we use a classifier or embedding similarity as context. (3) More context isn't always better because accuracy plateaus around $k \approx 7$. (4) NatureLM relies roughly 5x more on context quality than the audio itself. This is due to the fact that misleading context hurts accuracy ≈ 5 times as much as random audio. These findings show how text-conditioning of audio-LLMs deserves more attention rather than simply focusing on the audio side, which has historically gotten more focus.

In addition to our current study, it would be important to run this experiment on other audio-LLM models and other bioacoustic domains to see if RAG capabilities hold. Having a generalizable finding can lead to future work with RAG and audio-LLMs. Furthermore, since the model trusts context so heavily, a mechanism that determines how confident NatureLM should be in retrieved context is critical in understanding the model's capability. Adjusting this parameter and establishing a relationship to context quality allows for a more robust, fine-tuned model that could be made with future work. Finally, our duration-robustness finding, where longer clips did fine despite NatureLM seeing only the first 30 seconds, shows that NatureLM could have higher performance with longer clip duration, which could be tested. Ultimately, future work can help propel audioLLM with bioacoustics, helping with conservation of species and our planet.

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